

Week 6 Final Deliverable

Introduction

Following the successful completion of the Week 5 feasibility assessment, baseline machine learning models were developed to predict Emergency Severity Index (ESI) triage levels using the Yale Emergency Department dataset. The objective was not to build the most accurate model possible, but rather to establish a clinically defensible baseline that could be used for comparison with more advanced models in later stages of the project.

Three classifiers were evaluated: a stratified Dummy Classifier, Logistic Regression, and a Decision Tree. Model performance was assessed using accuracy, precision, recall, F1-score, and confusion matrices. Particular attention was given to the model's ability to correctly identify patients classified as **ESI Level 1**, representing the most critically ill patients.

Model Development

The cleaned dataset contained 55,121 emergency department encounters and 226 variables, including demographic information, physiological vital signs, and approximately 200 chief complaint indicators. After removing variables that would introduce data leakage, 209 predictor variables remained for modelling.

The dataset was divided using an 80/20 stratified train-test split with a fixed random seed of 42 to ensure reproducibility while maintaining the original ESI class distribution.

Three baseline models were trained:

- Dummy Classifier using stratified random prediction.
- Logistic Regression with balanced class weighting and standardised features.
- Decision Tree with a maximum depth of five to reduce overfitting and maintain interpretability.

Results

The Logistic Regression model achieved the strongest overall performance among the three models.

Model	Accuracy	Macro F1	Weighted F1
Dummy Classifier	0.375	0.204	0.375

Logistic Regression	0.563	0.412	0.596
Decision Tree	0.270	0.187	0.239

The Dummy Classifier provided a baseline for comparison by predicting classes according to their frequency within the training data. As expected, its performance was poor.

The Decision Tree performed worse than the Logistic Regression model across all evaluation metrics, suggesting that the shallow tree was unable to capture the complexity of the relationships within the dataset.

Logistic Regression achieved an overall accuracy of 56.3%, substantially outperforming the Dummy Classifier (37.5%) and the Decision Tree (27.0%). The model also produced the highest weighted F1-score (0.596), indicating better overall classification performance across the five ESI categories.

Primary Metric Justification

Although overall accuracy provides a useful summary of model performance, it is not the most clinically meaningful measure for emergency department triage because the dataset is highly imbalanced. Only 77 patients in the full dataset belonged to ESI Level 1, representing approximately 0.14% of all encounters.

For this reason, the primary evaluation metric is recall for ESI Level 1. Recall measures how many genuinely critically ill patients are successfully identified by the model.

The comparison between models illustrates this clearly:

- Dummy Classifier: 0.000
- Decision Tree: 0.500
- Logistic Regression: 0.625

The Logistic Regression model correctly identified approximately 62.5% of the patients requiring immediate life-saving intervention, compared with no successful identification by the Dummy Classifier.

From a clinical perspective, failing to recognise a critically ill patient could delay resuscitation or emergency treatment and lead to serious patient harm. In contrast, incorrectly assigning a stable patient to a higher priority category may increase resource utilisation but is generally considered the safer error.

Failure Mode Reflection

The most concerning failure mode is under-triage, where a patient requiring immediate intervention is classified into a less urgent ESI category.

The confusion matrices demonstrate that although Logistic Regression performs better than the other baseline models, it still misclassifies some patients across neighbouring ESI levels. Such errors could delay physician assessment, diagnostic investigations, and treatment for patients with life-threatening conditions.

Another limitation is the class imbalance within the dataset. Because ESI Level 1 cases are extremely uncommon, the model has relatively few examples from which to learn their characteristics. This imbalance partly explains why overall accuracy alone would provide an incomplete picture of model quality.

Finally, the dataset originates from a single United States emergency department. Differences in patient demographics, disease prevalence, healthcare resources, and clinical workflows mean that additional validation would be required before applying the model within Caribbean emergency departments.

Conclusion

This study successfully established baseline machine learning models for emergency department triage prediction. Logistic Regression consistently outperformed both the Dummy Classifier and the Decision Tree across overall accuracy, macro F1-score, weighted F1-score, and ESI Level 1 recall.

The results demonstrate that a relatively simple linear model provides a strong and interpretable baseline for future work. However, improvements remain necessary, particularly in identifying the small number of critically ill patients. Future work should investigate ensemble learning methods, feature selection, class balancing techniques, and external validation using Caribbean clinical data to improve both performance and generalisability.

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Github: <https://github.com/Amanda-miller0/carisurg-portfolio.git>